**SMASH - JSON CLONING HANDBOOK**
**How to turn a building concept into a map JSON that plays well**
**Companion to SMASH_MapJSON_Spec_and_Review.txt (the schema/importer reference)**
**Project: Demolition Contracts (FierceTigerGameJam) Revision 1 - 2026-08-29**

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WHO THIS IS FOR
    The AI or person producing map JSON from a building concept ("cloning" a
    building into the game). The spec document says what the importer ACCEPTS;
    this handbook says how to make what it accepts PLAY well: entry budgets,
    wall grouping that produces satisfying cascades, and the design patterns
    that keep ten buildings from looking like one box copied ten times.

THE ONE-PARAGRAPH VERSION
    Plan on a 0.25 m grid. Build a hollow shell with real openings. Assign
    every structural cell to a NAMED WALL of 4-30 cells, one material per
    wall, split by face and by storey. Merge even horizontal brick runs into
    brick_2x1. Keep every body supported. Validate, voxel-render, compare to
    the concept, ship JSON + render + manifest together.
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## 1. THE CLONING PIPELINE - SEVEN STEPS, IN ORDER

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STEP 1 - SILHOUETTE FIRST
    Decide the massing before any block exists: footprint (width x depth in
    cells), height in rows, and the one feature that makes this building
    recognisable at a glance (a tower, a setback, a gate, a glass belt).
    Rule: if the silhouette reads as a plain box, add exactly one move from
    section 5 before continuing. Draw or render the concept AT CELL
    RESOLUTION - one brick in the concept = one 0.25 m cell - so the concept
    can never promise detail the JSON cannot deliver.

STEP 2 - GRID PLAN
    width  = widest x extent used, 4-16 cells
    height = tallest row used + 1, 3-14 cells
    layers = depth slices, 2-12; a free-standing building wants depth close
             to its width (a 10-wide building with 4 layers is a thick wall,
             not a house)
    cellSize = 0.25, layerDepth = 0.25, always. They are spacing, not scale.

STEP 3 - SHELL, HOLLOW
    Author the perimeter walls one cell thick and leave the core EMPTY.
    Interior cells are only spent on: floor slabs between storeys (one row,
    they make a collapse "pancake" in stages), interior columns holding a
    slab, and courtyard floors. A solid-core building wastes half its budget
    on cells the player never sees and never enjoys breaking.

STEP 4 - OPENINGS ARE HOLES, NOT DECORATION
    Doorway: 2 wide x 3 tall gap at row 0, bridged by a lintel row that
    belongs to the wall beside it (a lintel with its own wall id has no
    footing and hangs forever - known failure, see spec 3.1).
    Windows: 1x1 or 2x2 gaps, or glass-filled cells when they should shatter.
    Crenellations: alternate filled/empty on the top row.
    Every hole is free budget AND better gameplay - shots pass through.

STEP 5 - MATERIALS WITH INTENT
    brick    = the body of the building (3 HP, mass 1.2)
    glass    = what the player pops for fun (1 HP, mass 0.6) - windows,
               curtain belts, skylights
    concrete = the skeleton that resists (6 HP, mass 2.0) - corner piers,
               ground frames, lintel bands
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Progression across a campaign: pure brick -> brick + glass accents -> concrete frame + brick infill + glass. Never mix materials INSIDE one wall id: the engine takes the first block's material for the whole wall, so a mixed wall sounds and scores wrong.

STEP 6 - WALL GROUPING (the step that makes or breaks the feel - see sec. 3)

STEP 7 - VERIFY AND SHIP

Run the validator (schema, bounds, overlap, support-per-body, wall connectivity, single-material walls). Voxel-render the JSON and put it beside the concept - if they differ, the JSON is the one that ships, so fix the JSON or re-render the concept. Deliver: the .json, the render, and a manifest line (id, grid, entries, walls, loose bodies, materials).

## 2. ENTRY OPTIMISATION - FEWER OBJECTS, SAME BUILDING

Every JSON entry becomes at least one GameObject. Two mechanical reductions are always available:

2.1 brick_2x1 MERGING

Any two horizontally adjacent brick_1x1 cells at the same row and layer, assigned to the SAME wall, can be one brick_2x1 (rotation 0, positioned at the left cell). Measured on the current pack this cut 15-22% of entries per map (96->77, 254->200, 306->262).
The trade, stated honestly: two 1x1 bricks are 6 HP, one 2x1 is 5 HP, so a merged wall is a few percent weaker in total; masses are identical (2.4). Do not merge: glass (shatters per pane), deliberate loose accent blocks, or across a wall boundary.

2.2 HOLES OVER FILLER

The cheapest block is the one you do not author. Openings, hollow cores and courtyards do more for both frame rate and fun than any merging.

Budgets (from the spec, restated): tutorial 50-120 entries, mid 120-220, hard 220-350, boss 280-400; loose bodies <= 150 on the opening frame; and remember merging lowers the STANDING cost only - a knocked wall spawns its blocks back, so the total cell count still bounds the worst frame.

## 3. WALL GROUPING FOR CASCADES - THE ASMR ALGORITHM

WHY WALL SIZE IS THE FEEL DIAL

| | |
|---|---|
| One wall per building | -> the whole house topples as a single slab: one thud, no texture, and a huge HP pool the player chips at with no visible progress. (This was the pack's one systematic fault: 96-236 cells in one wall on five of nine maps.) |
| Everything loose | -> hundreds of rigidbodies, mush on Android, and destruction reads as static. |
| MANY MEDIUM WALLS | -> each hit brings down a satisfying chunk, the chunk bursts into blocks on landing, and the storey above follows a beat later. That staggered thud-thud-crunch IS the ASMR. |

THE NUMBERS

Wall size sweet spot: 4-24 cells. Hard bounds 2 (engine minimum) and 30. A mid-size map should end up with 10-25 walls plus a handful of loose accent blocks (5-10% of entries).

THE SPLITTING ALGORITHM (deterministic, scriptable - this is exactly what produced the current optimised pack):

1. Classify every cell by FACE: left (x = min), right (x = max), front (z = min), back (z = max), roof (y = max), floor (y = 0 interior),

core (the rest). Side faces claim the corners so each face is a clean slab.
2. Within each (material, face) group, cut vertical faces into STOREY BANDS of 3 rows (y / 3). Bands are what makes a building fall floor by floor instead of all at once.
3. Within each band, split into CONNECTED components (6-neighbour) - the engine splits disconnected walls itself, noisily; do it first.
4. Any component over 30 cells is halved along x until it fits.
5. Components of 1 cell become loose blocks.
6. FOOTING REPAIR: any wall whose every cell rests only on itself or on air is merged into a touching neighbour wall (prefer the one below). Run to fixpoint. This is what keeps a split roof from hovering.

Wall ids: face_material_band_index (front_brick_b1_4) - readable in the hierarchy, unique by construction, ordinal-compared by the engine.

WHAT NOT TO GROUP

Glass: one wall per window pane group of 2-6, or fully loose for big shatter moments. Never fold glass into a structural wall.
Crenellations, ledge tips, chimney caps: loose, so they rain down individually when the wall under them goes.

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## 4. DESTRUCTION FEEL - JSON-SIDE DIRECTIVES

The engine already staggers activation (kinematic until hit, support cascade upward, walls burst into blocks on landing). The JSON decides whether that machinery gets material to perform with:

D1 STAGED COLLAPSE. Storey bands (sec. 3) + one interior floor slab per storey = the top floor pancakes onto the slab, the slab breaks the floor below, three distinct crunches from one shot.

D2 DOMINO PILLARS. A heavy roof or slab carried by 2-3 slim pillars (1x2 or 1x3 concrete walls): shoot one pillar, the slab tips, the slab wipes the far wall. Author the pillars as separate walls so one shot fells exactly one.

D3 GLASS RAIN. A continuous glass belt or curtain ABOVE head height, held up by the wall beneath: when that wall band falls, every pane above it releases in the same second. (map_004's belt is the reference; push it to full curtain walls on later levels.)

D4 CROWN DEBRIS. Loose 1x1s on the very top (crenellations, chimney pots, parapet caps): the finale of any collapse is single bricks bouncing down the rubble. Cheap, reads beautifully.

D5 ONE WEAK SEAM. Give every building a visible "aim here" line - a glass band across the waist, a brick band inside a concrete frame - so the player discovers the collapse rather than sanding the building down.

D6 SOUND IDENTITY. One material per wall (sec. 1.5) is also an audio rule: each falling chunk should play as brick OR glass OR concrete. (Engine side, for the team: map the WhackStack per-material hit and crack sounds onto BreakableWall break-up and block landings, add a low thump scaled by wall cell count, and a short 0.2 s slow-mo when a wall of 15+ cells breaks - those three close the ASMR loop.)

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## 5. ARCHITECTURE VARIETY - TWELVE MOVES THAT BREAK THE BOX

Pick ONE primary move per building (two for boss maps). All of them are plain cell geometry - no new engine features needed:

A1 STEPPED TOWER (setback high-rise, the way real city towers step back as they climb): footprint shrinks by 1-2 cells every 3 rows. Every ledge is a debris shelf during collapse, and the profile reads as a building under construction rather than a box.

A2 CORNER TOWER: one corner runs 3-4 rows taller than the body, with

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    its own crenellated top. Fells sideways INTO the building - D2.
A3  L / T / U FOOTPRINT: drop a quadrant of the plan, or push a wing
    forward in z. Costs nothing; kills the box silhouette instantly.
A4  GATEHOUSE: two short towers flanking a tall arched gap, bridged by
    a lintel band - the bridge falling into the gap is a set piece.
A5  TWIN TOWERS + SKYBRIDGE: two 3x3 towers, a 1-row slab spanning
    them at mid-height. Shoot either tower, the bridge goes.
A6  GLASS WAIST: one full storey of glass between two brick storeys -
    the building visibly "sits" on its weakest floor (D5 built in).
A7  CONCRETE EXOSKELETON: concrete frame (corners + lintel bands),
    brick infill panels. Player learns to punch panels, then frame.
A8  BUTTRESSES: 1-cell-deep fins on the outer face every 3 cells.
    Texture in silhouette and extra loose debris.
A9  CANTILEVER BALCONY: a 2-cell slab jutting out, propped by one
    column. Classic one-shot domino.
A10 CHIMNEYS / ROOF FURNITURE: 1x2 stacks on the roof, loose caps.
A11 RUIN VARIANT: bite 10-20% of the shell off one top corner with
    real gaps and a few loose bricks at the break line - reads as
    battle-damaged, not as floating-block noise.
A12 COURTYARD + SITE SHED: ring building with an interior court and
    one small free-standing structure inside it (site office, storage
    shed) as a bonus target.

ASYMMETRY RULE: at most one axis of symmetry per building. Perfectly
symmetric buildings read as boxes whatever else they do.
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## 6. FINAL CHECKLIST (extends the spec's section 9)

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[ ] silhouette uses one move from section 5; concept drawn at cell scale
[ ] hollow core; interior cells only for slabs/columns/courts
[ ] every structural cell in a named wall of 2-30 cells (target 4-24)
[ ] vertical faces banded by storey; disconnected walls pre-split
[ ] every wall footed (repair pass ran); no floating loose block
[ ] one material per wall; glass never inside a structural wall
[ ] even brick runs merged to brick_2x1 within walls
[ ] 10-25 walls, loose <= 10% of entries, entries within the level band
[ ] at least one of D1-D5 present by design
[ ] validator: 0 errors; voxel render matches concept side by side
[ ] delivered: .json + render + manifest line
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**END**